

LAKE VYRNWY, United Kingdom

WMO: 034100

Lat: 52.7572N

Lon: 3.4653W

Elev: 359

StdP: 97.09

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.8	-2.6	-6.2	2.3	-1.4	-5.0	2.6	-1.1	23.7	5.9	19.5	7.0	3.7	70	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.6	22.6	17.5	20.6	16.0	18.9	15.1	17.9	20.9	16.7	19.4	15.8	18.1	3.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.5	19.0	15.7	11.7	17.5	14.9	11.0	16.6	51.9	21.0	48.2	19.5	45.5	18.2	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	12.8	10.9	DB	-5.4	24.7	2.0	2.3	-6.9	26.3	-8.0	27.6	-9.1	28.9	-10.6	30.6	(4)
(5)			WB	-5.8	20.0	1.9	2.1	-7.1	21.6	-8.3	22.8	-9.3	24.0	-10.7	25.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	3.3	3.6	4.8	7.0	9.7	12.5	14.2	13.9	12.1	9.2	6.1	4.0	(6)
(7)		DBStd	4.77	2.98	2.89	2.87	2.85	2.78	2.50	2.48	2.12	2.14	2.51	2.70	3.22	(7)
(8)		HDD10.0	1046	208	180	163	97	39	4	0	0	6	44	118	186	(8)
(9)		HDD18.3	3633	467	413	421	340	267	175	130	139	188	283	367	444	(9)
(10)		CDD10.0	458	0	0	1	7	31	80	132	121	67	19	2	0	(10)
(11)		CDD18.3	6	0	0	0	0	0	1	3	2	0	0	0	0	(11)
(12)		CDH23.3	30	0	0	0	0	0	4	17	8	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	4.4	5.3	5.3	4.6	4.0	4.1	3.7	3.7	3.8	4.5	4.5	5.2	(14)	
(15)	Precipitation	PrecAvg	1353	159	126	94	84	89	82	84	85	95	140	148	168	(15)
(16)		PrecMax	1772	271	310	191	165	161	187	188	135	199	282	362	366	(16)
(17)		PrecMin	954	9	24	21	14	32	14	31	15	6	28	68	34	(17)
(18)		PrecStd	204	76	72	43	44	37	46	39	30	48	71	74	87	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.5	11.8	13.8	18.5	21.4	23.8	25.3	24.5	21.1	16.7	13.0	11.5	(19)
(20)			MCWB	9.4	9.0	9.5	12.3	16.0	17.4	18.4	17.7	15.9	14.1	12.1	10.8	(20)
(21)		2%	DB	9.4	9.7	11.4	15.3	18.6	20.9	23.2	20.9	18.5	14.8	12.0	10.6	(21)
(22)			MCWB	8.6	8.5	8.6	10.6	14.0	16.3	18.3	16.2	15.3	13.5	11.3	10.0	(22)
(23)		5%	DB	8.5	8.7	10.0	13.1	16.4	19.0	21.1	19.0	16.8	13.8	11.1	9.6	(23)
(24)			MCWB	7.7	7.6	8.1	9.4	12.6	14.9	16.7	15.4	14.3	12.5	10.2	9.0	(24)
(25)		10%	DB	7.6	7.7	8.8	11.3	14.5	17.2	18.9	17.4	15.6	12.9	10.1	8.6	(25)
(26)			MCWB	6.8	6.7	7.4	8.5	11.3	13.8	15.4	14.8	13.6	11.7	9.2	7.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.9	10.5	12.8	16.5	19.6	22.8	18.7	17.6	15.0	12.5	11.1	(27)
(28)			MCDWB	10.4	10.9	12.2	16.7	20.1	21.7	23.2	21.8	19.2	15.4	12.9	11.3	(28)
(29)		2%	WB	8.7	8.8	9.3	11.2	14.7	16.8	18.3	17.0	16.0	13.8	11.5	10.1	(29)
(30)			MCDWB	9.2	9.6	10.7	14.1	17.8	19.6	21.5	19.2	17.6	14.7	11.9	10.5	(30)
(31)		5%	WB	7.9	7.9	8.5	10.1	13.2	15.4	16.9	16.2	15.0	12.9	10.4	9.1	(31)
(32)			MCDWB	8.5	8.5	9.7	12.3	15.6	18.1	20.4	18.2	16.3	13.7	10.9	9.5	(32)
(33)		10%	WB	6.9	6.9	7.6	9.0	12.0	14.2	15.8	15.4	14.0	11.9	9.3	7.9	(33)
(34)			MCDWB	7.6	7.7	8.7	11.0	14.0	16.6	18.3	17.0	15.3	12.6	9.9	8.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.3	5.2	6.4	6.7	6.7	6.6	6.0	5.8	4.7	4.1	4.0	(35)
(36)			MCDBR	4.9	5.4	7.0	9.1	9.6	10.0	10.3	8.3	7.7	5.6	4.7	4.9	(36)
(37)			MCWBR	5.1	4.8	5.1	5.6	5.9	6.5	6.1	4.8	5.3	4.6	4.7	5.1	(37)
(38)		5% WB	MCDWR	4.7	5.0	6.2	7.8	8.6	8.8	9.0	6.7	6.9	5.3	4.7	4.9	(38)
(39)			MCWBR	5.1	4.8	5.0	5.2	5.7	6.4	6.0	4.6	5.5	4.9	4.9	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.326	0.355	0.377	0.382	0.375	0.374	0.359	0.350	0.335	0.312	0.302	(40)	
(41)		taud	2.331	2.386	2.312	2.272	2.297	2.341	2.347	2.404	2.430	2.439	2.392	2.298	(41)	
(42)		Ebn at Noon	669	783	828	854	868	877	870	865	829	762	668	603	(42)	
(43)		Edh at Noon	61	77	101	119	122	118	115	103	89	73	57	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.70	1.35	2.48	3.83	4.78	5.07	4.75	3.89	2.90	1.62	0.84	0.53	(44)	
(45)		RadStd	0.06	0.15	0.27	0.42	0.30	0.53	0.51	0.29	0.23	0.15	0.09	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page